

L04. The Linear Quadratic Regulator (LQR)

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1 Introduction

In the previous lecture, we developed the dynamic programming recursion, expressed through Bellman's equation. The idea is to work recursively backward in time: once we know the optimal cost-to-go at the next time step, we can choose the best action at the current step by accounting for both its immediate cost and its effect on future costs. Starting from the terminal cost, we repeat this process backward to obtain an optimal policy.

The *Linear Quadratic Regulator (LQR)* uses these same principles. In a sense, we are already done: Bellman's equation tells us how to solve the problem. However, we can use LQR to really dig deeper into what is going on.

Aside

Please note that we use x to represent the state and u to represent the control in this lecture, in accordance with control theory.

2 Linear Quadratic Regulator

Linear models are widely used in dynamic control problems. The motivation is that many smooth systems look approximately linear when we consider small changes around an operating point: think of how a derivative gives a local linear approximation to a function. Even when the full dynamics are nonlinear, a linear model can therefore be useful near the point where we want to operate.

For example, consider the vertical motion of a helicopter. Let $h(t)$ be its height, $F(t)$ the upward thrust, and m its mass. Ignoring other forces, Newton's law gives

$$m\ddot{h}(t) = F(t) - mg, \quad \text{or equivalently} \quad \ddot{h}(t) = \frac{F(t)}{m} - g.$$

We would like to control the thrust so that the helicopter reaches a desired height h_D and stays there.

At hover, the thrust balances gravity: $F(t) = mg$. Thus, it is helpful to define the control input as the change in thrust relative to this hover value:

$$u(t) := F(t) - mg.$$

Then the dynamics become simply $\ddot{h}(t) = u(t)/m$. If we collect height and vertical velocity into the state,

$$\bar{x}(t) = \begin{bmatrix} h(t) \\ \dot{h}(t) \end{bmatrix},$$

we can write these dynamics as a linear system:

$$\dot{\bar{x}}(t) = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_A \bar{x}(t) + \underbrace{\begin{bmatrix} 0 \\ 1/m \end{bmatrix}}_B u(t).$$

We can also express this motion using discrete time steps. For a small sampling interval Δt , we get approximately

$$\bar{x}_{k+1} \approx \bar{x}_k + \Delta t \dot{\bar{x}}(k\Delta t).$$

or exactly

$$h_{k+1} = h_k + \Delta t \dot{h}_k + \frac{\Delta t^2}{2m} u_k, \quad \dot{h}_{k+1} = \dot{h}_k + \frac{\Delta t}{m} u_k,$$

which we can rewrite as

$$\bar{x}_{k+1} = \underbrace{\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}}_{A_d} \bar{x}_k + \underbrace{\begin{bmatrix} \Delta t^2/(2m) \\ \Delta t/m \end{bmatrix}}_{B_d} u_k.$$

Now that we know why we care about linear models, how do we formulate the general control problem?

2.1 Setting up the Problem

We consider a finite time horizon $t = 0, 1, \dots, T$. First, we model the state transitions in a linear model (the linear part of LQR):

$$x_{t+1} = A_t x_t + B_t u_t + w_t, \quad t = 0, \dots, T-1.$$

The matrix A_t describes how the current state evolves, while B_t describes how the control affects the next state. The term $w_t \in \mathbb{R}^n$ represents zero-mean i.i.d. noise.

Next, we specify what we want to minimize. At each time step, we assign the quadratic cost (the quadratic part of LQR)

$$c_t(x_t, u_t) = \underbrace{x_t^\top Q_t x_t}_{\text{state penalty}} + \underbrace{u_t^\top R_t u_t}_{\text{control penalty}}.$$

At the final time, we also assign a terminal cost:

$$c_T(x_T) = x_T^\top Q_T x_T.$$

What does each part of the cost mean?

The first term penalizes deviations from the desired state, with Q_t determining which deviations matter most. For the helicopter, this would mean using $x_t = \bar{x}_t - [h_D, 0]^\top = [h_t - h_D, \dot{h}_t]^\top$: we want to maintain the desired height and have zero vertical velocity.

The second term penalizes the amount of control used, with R_t determining how costly that effort is. For the helicopter, this means penalizing the magnitude of our control input $u(t) := F(t) - mg$. In systems with multiple control inputs (non-scalar u), R_t can also assign different costs to different control channels.

2.2 Understanding the Linear Problem

We first try to gain some intuition about the problem. Before choosing the optimal controls, we examine how a given control sequence affects the state. We can solve the linear dynamics explicitly by repeated substitution (this expansion is not needed for the dynamic programming section, but will be useful later for reachability).

Starting from

$$x_{t+1} = A_t x_t + B_t u_t + w_t,$$

we get

$$\begin{aligned} x_1 &= A_0 x_0 + B_0 u_0 + w_0, \\ x_2 &= A_1 x_1 + B_1 u_1 + w_1 \\ &= A_1 A_0 x_0 + A_1 B_0 u_0 + A_1 w_0 + B_1 u_1 + w_1, \\ x_3 &= A_2 x_2 + B_2 u_2 + w_2 \\ &= A_2 A_1 A_0 x_0 + A_2 A_1 B_0 u_0 + A_2 B_1 u_1 + B_2 u_2 \\ &\quad + A_2 A_1 w_0 + A_2 w_1 + w_2. \end{aligned}$$

To write this compactly, define the state transition matrix

$$\Phi(t, \tau) := A_{t-1} A_{t-2} \cdots A_\tau, \quad t > \tau, \quad \Phi(t, t) := I.$$

We then get the closed form solution

$$\begin{aligned} x_t &= \underbrace{\Phi(t, 0) x_0}_{\text{initial state}} + \underbrace{\sum_{\tau=0}^{t-1} \Phi(t, \tau+1) B_\tau u_\tau}_{\text{controls}} + \underbrace{\sum_{\tau=0}^{t-1} \Phi(t, \tau+1) w_\tau}_{\text{noise}} \\ &= \Phi(t, 0) x_0 + \sum_{\tau=0}^{t-1} \Phi(t, \tau+1) (B_\tau u_\tau + w_\tau). \end{aligned}$$

In the *linear time-invariant (LTI)* case, the matrices do not change with time: $A_t = A$ and $B_t = B$. The products of transition matrices then become powers of A :

$$\Phi(t, \tau) = A^{t-\tau}.$$

Thus,

$$x_t = A^t x_0 + \sum_{\tau=0}^{t-1} A^{t-1-\tau} B u_\tau + \sum_{\tau=0}^{t-1} A^{t-1-\tau} w_\tau.$$

This gives the convolution structure: the contribution of each past input depends on how many time steps have passed since it was applied.

2.3 Formalizing the Objective

However, we want to find the optimal policy: what is the best control at each time step? We can solve this problem by relating back to dynamic programming and reinforcement learning.

Our objective is to choose a policy π , with $u_t = \pi_t(x_t)$, that minimizes the expected total cost:

$$\min_{\pi} \mathbb{E} \left[\sum_{t=0}^{T-1} c_t(x_t, u_t) + c_T(x_T) \middle| x_0 \right].$$

Let $V_t(x)$ denote the optimal expected cost-to-go from state x at time t . Bellman's equation gives

$$V_t(x) = \min_u \{ x^\top Q_t x + u^\top R_t u + \mathbb{E}_{w_t} [V_{t+1}(A_t x + B_t u + w_t)] \},$$

recurring backwards from

$$V_T(x) = x^\top Q_T x.$$

Similar to last lecture, we can continue at $T-1$:

$$\begin{aligned} V_{T-1}(x) &= \min_u \left\{ x^\top Q_{T-1} x + u^\top R_{T-1} u + \mathbb{E}_{w_{T-1}} [x_T^\top Q_T x_T] \right\} \\ &= \min_u \left\{ x^\top Q_{T-1} x + u^\top R_{T-1} u \right. \\ &\quad \left. + \mathbb{E}_{w_{T-1}} \left[(A_{T-1} x + B_{T-1} u + w_{T-1})^\top Q_T (A_{T-1} x + B_{T-1} u + w_{T-1}) \right] \right\}. \end{aligned}$$

However, unlike last lecture, we can keep going! Expanding the quadratic, we get:

$$\begin{aligned} V_{T-1}(x) &= \min_u \left\{ x^\top Q_{T-1} x + u^\top R_{T-1} u \right. \\ &\quad + (A_{T-1} x + B_{T-1} u)^\top Q_T (A_{T-1} x + B_{T-1} u) \\ &\quad \left. + 2(A_{T-1} x + B_{T-1} u)^\top Q_T \underbrace{\mathbb{E}[w_{T-1}]}_{=0} + \mathbb{E}[w_{T-1}^\top Q_T w_{T-1}] \right\} \\ &= \min_u \left\{ x^\top Q_{T-1} x + u^\top R_{T-1} u \right. \\ &\quad + (A_{T-1} x + B_{T-1} u)^\top Q_T (A_{T-1} x + B_{T-1} u) \\ &\quad \left. + \mathbb{E}[w_{T-1}^\top Q_T w_{T-1}] \right\} \\ &= \min_u \left\{ x^\top (Q_{T-1} + A_{T-1}^\top Q_T A_{T-1}) x \right. \\ &\quad + u^\top (R_{T-1} + B_{T-1}^\top Q_T B_{T-1}) u + 2x^\top A_{T-1}^\top Q_T B_{T-1} u \\ &\quad \left. + \mathbb{E}[w_{T-1}^\top Q_T w_{T-1}] \right\}. \end{aligned}$$

To find our optimal control, we can drop the last term as they are not dependent on u (and the first, but we will keep as it is informative for our recursion). We can write

$$f(x, u) = x^\top F x + u^\top G u + 2x^\top H^\top u,$$

where

$$\begin{aligned} F &= Q_{T-1} + A_{T-1}^\top Q_T A_{T-1}, \\ G &= R_{T-1} + B_{T-1}^\top Q_T B_{T-1}, \\ H &= B_{T-1}^\top Q_T A_{T-1}. \end{aligned}$$

Note F and G are symmetric.

Aside

Why can we assume symmetry? For any real square matrix S ,

$$x^\top S x = \frac{1}{2}(x^\top S x + x^\top S^\top x) = x^\top \left(\frac{S + S^\top}{2} \right) x.$$

Thus, replacing a matrix by its symmetric part leaves its quadratic form unchanged.

2.4 Why positive semidefinite

Similar to last lecture, we would like the optimization in to be convex in order to solve it. But are all quadratic functions convex?

No: for example, $f(x_1, x_2) = x_1^2 - x_2^2$ has a saddle point and is unbounded below along the x_2 direction. Thus, having a quadratic cost is not enough.

Thus, for LQR, we have to make additional assumptions. Those assumptions are that Q_t is positive semidefinite and R_t is positive definite.

Aside

A real symmetric matrix M is positive semidefinite if

$$z^\top M z \geq 0 \quad \text{for every } z.$$

It is positive definite if

$$z^\top M z > 0 \quad \text{for every } z \neq 0.$$

Equivalently, its eigenvalues are all nonnegative or all strictly positive, respectively. Please see 18.06 Material for review.

First, these assumptions ensure that our costs are nonnegative:

$$x^\top Q_t x \geq 0, \quad u^\top R_t u \geq 0.$$

Second, these assumptions give us the convexity we need. We can rewrite $f(x, u)$ as

$$f(x, u) = \begin{bmatrix} x \\ u \end{bmatrix}^\top \underbrace{\begin{bmatrix} F & H^\top \\ H & G \end{bmatrix}}_M \begin{bmatrix} x \\ u \end{bmatrix}.$$

The function is jointly convex in (x, u) if and only if $M \succeq 0$, or positive semidefinite. This is true when Q_T and R_T are positive semidefinite, as

$$M = \begin{bmatrix} Q_{T-1} & 0 \\ 0 & R_{T-1} \end{bmatrix} + \begin{bmatrix} A_{T-1} & B_{T-1} \end{bmatrix}^\top Q_T \begin{bmatrix} A_{T-1} & B_{T-1} \end{bmatrix}.$$

The first term is positive semidefinite because its quadratic form is $x^\top Q_{T-1} x + u^\top R_{T-1} u$, a sum of non-negative terms. The second is positive semidefinite as $z^\top C^\top Q_T C z = (Cz)^\top Q_T (Cz) \geq 0$. Thus, M is positive semidefinite, and the function f jointly convex in the state and control. Minimizing over u preserves convexity in x , giving us a convex value function as desired!

Aside

We can prove that positive semidefinite implies convexity. One is using the Hessian (think second derivative). Writing $z = [x^\top, u^\top]^\top$ and using the symmetry of M , we have

$$f(z) = z^\top M z, \quad \nabla f(z) = 2Mz, \quad \nabla^2 f(z) = 2M.$$

Thus, the Hessian is positive semidefinite exactly when M is positive semidefinite (with strict convexity when M is definite).

The other way is more directly. For any vectors z_1, z_2 and $\theta \in [0, 1]$, we have

$$\begin{aligned} f(\theta z_1 + (1 - \theta)z_2) &= \theta^2 z_1^\top M z_1 + 2\theta(1 - \theta)z_1^\top M z_2 + (1 - \theta)^2 z_2^\top M z_2 \\ &= \theta f(z_1) + (1 - \theta)f(z_2) \\ &\quad - \theta(1 - \theta)(z_1 - z_2)^\top M (z_1 - z_2). \end{aligned}$$

Since M is positive semidefinite, $(z_1 - z_2)^\top M (z_1 - z_2) \geq 0$. We are therefore subtracting a nonnegative quantity, giving

$$f(\theta z_1 + (1 - \theta)z_2) \leq \theta f(z_1) + (1 - \theta)f(z_2),$$

which is the definition of convexity.

2.5 Why positive definite

You might note that we require R_t to be positive definite, rather than only positive semidefinite. Why do we ask for this stronger condition?

If R_t were only positive semidefinite, it could have a nonzero vector v in its null space: $R_t v = 0$. A control in this direction would have no direct cost, regardless of its magnitude:

$$u = \alpha v \implies u^\top R_t u = \alpha^2 v^\top R_t v = 0.$$

For example, if

$$R_t = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

then $u^\top R_t u = u_1^2$: the second control input receives no direct penalty. This does not necessarily make it cost-free overall, since it may affect future states and their costs. However, if the future-state cost also leaves this direction unpenalized, the minimizing control may not be unique.

Your next question might be why do we allow Q_t to be only positive semidefinite? Here, we may not need to penalize every state direction directly. Some state coordinates can have zero weight while the minimization over the control still has a unique solution.

2.6 Solving the Objective

Now that we have a convex value function, we can solve for the optimal control. Recall the quadratic function

$$f(x, u) = x^\top F x + u^\top G u + 2x^\top H^\top u.$$

For a fixed state x , we differentiate with respect to u and set the gradient equal to zero:

$$\nabla_u f(x, u) = 2Gu + 2H^\top x = 0.$$

Since $G \succ 0$, the function is strictly convex in u , so this stationary point is the unique minimum. Positive definiteness also guarantees that G is invertible, giving

$$u^*(x) = -G^{-1}Hx.$$

What is the structure of our optimal control? It is a linear function of the state! We did not assume that the policy had to be linear; this follows from solving the quadratic optimization problem.

Now, let us put this control back into the objective. Since G and G^{-1} are symmetric,

$$\begin{aligned} f(x, u^*(x)) &= x^\top Fx + (-G^{-1}Hx)^\top G(-G^{-1}Hx) + 2x^\top H^\top (-G^{-1}Hx) \\ &= x^\top Fx + x^\top H^\top G^{-1}Hx - 2x^\top H^\top G^{-1}Hx \\ &= x^\top (F - H^\top G^{-1}H)x. \end{aligned}$$

Why is this significant? After minimizing over u , we are left with a quadratic function of x . This means the same structure can repeat at the previous time step.

Returning to our optimal control problem, at time $T-1$ we have

$$\begin{aligned} F &= Q_{T-1} + A_{T-1}^\top Q_T A_{T-1}, \\ G &= R_{T-1} + B_{T-1}^\top Q_T B_{T-1}, \\ H &= B_{T-1}^\top Q_T A_{T-1}. \end{aligned}$$

Here, $G \succ 0$ because $R_{T-1} \succ 0$ and $B_{T-1}^\top Q_T B_{T-1} \succeq 0$. Thus, our optimal control is

$$u_{T-1}^*(x) = -(R_{T-1} + B_{T-1}^\top Q_T B_{T-1})^{-1} B_{T-1}^\top Q_T A_{T-1} x.$$

Substituting this back into the Bellman objective, and retaining the noise term, gives

$$V_{T-1}(x) = x^\top K_{T-1} x + \mathbb{E}[w_{T-1}^\top Q_T w_{T-1}],$$

where

$$\begin{aligned} K_{T-1} &= Q_{T-1} + A_{T-1}^\top Q_T A_{T-1} \\ &\quad - A_{T-1}^\top Q_T B_{T-1} (R_{T-1} + B_{T-1}^\top Q_T B_{T-1})^{-1} B_{T-1}^\top Q_T A_{T-1}. \end{aligned}$$

So the optimal value is quadratic in the state, plus a constant that does not depend on the state.

We can now continue the iteration. At time $T-2$, the next value function has the same form as before, with K_{T-1} taking the place of Q_T and an additional constant. Thus, the same calculation applies.

More generally, suppose

$$V_{t+1}(x) = x^\top K_{t+1} x + d_{t+1},$$

where d_{t+1} is independent of x . Substituting into Bellman's equation and taking the expectation over the zero-mean noise gives

$$\begin{aligned} V_t(x) &= d_{t+1} + \mathbb{E}[w_t^\top K_{t+1} w_t] \\ &\quad + \min_u \left\{ x^\top F_t x + u^\top G_t u + 2x^\top H_t^\top u \right\}, \end{aligned}$$

where

$$\begin{aligned} F_t &= Q_t + A_t^\top K_{t+1} A_t, \\ G_t &= R_t + B_t^\top K_{t+1} B_t, \\ H_t &= B_t^\top K_{t+1} A_t. \end{aligned}$$

The noise cross terms vanish just as they did at $T - 1$. Here, we assume the fresh noise is independent of the current state and control and has finite second moments.

Applying our quadratic minimization formula again,

$$u_t^*(x) = -G_t^{-1}H_t x = -L_t x, \quad L_t := G_t^{-1}B_t^\top K_{t+1}A_t.$$

The value function is

$$V_t(x) = x^\top K_t x + d_t,$$

with

$$\begin{aligned} K_t &= F_t - H_t^\top G_t^{-1} H_t \\ &= Q_t + A_t^\top K_{t+1} A_t - A_t^\top K_{t+1} B_t G_t^{-1} B_t^\top K_{t+1} A_t, \end{aligned}$$

and

$$d_t = d_{t+1} + \mathbb{E}[w_t^\top K_{t+1} w_t].$$

The terminal conditions are

$$K_T = Q_T, \quad d_T = 0.$$

The recursion for K_t is the discrete-time *Riccati equation*. Equivalently,

$$K_t = Q_t + A_t^\top (K_{t+1} - K_{t+1} B_t G_t^{-1} B_t^\top K_{t+1}) A_t, \quad G_t = R_t + B_t^\top K_{t+1} B_t.$$

At each step, K_t remains positive semidefinite: $x^\top K_t x$ is the minimum of the nonnegative quadratic

$$x^\top Q_t x + u^\top R_t u + (A_t x + B_t u)^\top K_{t+1} (A_t x + B_t u).$$

Starting from $K_T = Q_T \succeq 0$, this also guarantees that $G_t \succ 0$ at every step, so we can continue the recursion.

This is the same backward dynamic programming argument as before, but now we can carry it out explicitly. We compute the matrices backward from time T , then apply $u_t = -L_t x_t$ as the system evolves forward. The optimal policy is linear, and the value function remains quadratic up to an additive constant at every time step.

3 Infinite Time Horizon

So far, we have assumed a finite horizon $t = 0, \dots, T$. What happens as T approaches infinity? For an LTI system with fixed A, B, Q, R , we would like the Riccati matrices to approach a finite limit $\bar{K}$, giving a stationary linear control law. At first glance, this may be obvious as we are minimizing at every step.

However, this may not be true. An unstable state direction may be impossible to control, or it may grow without being penalized by Q . Thus, we need to investigate both convergence of the Riccati recursion and stability of the resulting system.

Two useful properties are *reachability*: can we move the state where we want? And *observability*: can a nonzero state remain invisible to the output, and hence potentially to the cost?

3.1 Reachability

Consider

$$x_{k+1} = Ax_k + Bu_k.$$

Starting from any state x_0 , can we choose controls that reach any desired state x_f in finite time?

Using our earlier closed-form solution,

$$x_N = A^N x_0 + \underbrace{\begin{bmatrix} B & AB & \cdots & A^{N-1}B \end{bmatrix}}_{\mathcal{R}_N} \begin{bmatrix} u_{N-1} \\ u_{N-2} \\ \vdots \\ u_0 \end{bmatrix}.$$

To reach every x_f , the columns of $\mathcal{R}_N$ must span the state space. For an n -dimensional LTI system, the reachability condition then becomes

$$\text{rank} \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} = n.$$

Essentially, our controls must be able to influence every state direction, either immediately or through the dynamics over time. When this does not hold, some states cannot be reached. If an unreachable part of the system is unstable, no control we apply can prevent the growth of that state component.

3.2 Observability

Now consider an output $y_k = Cx_k$. Observability asks whether we can determine the state from its outputs over time. Under zero input, the system is observable if the only initial state satisfying

$$y_k = CA^k x_0 = 0 \quad \text{for every } k$$

is $x_0 = 0$.

Remember that we do not require Q to be positive definite. Writing $Q = C^\top C$, our state cost becomes

$$x_k^\top Q x_k = \|Cx_k\|^2 = \|y_k\|^2.$$

Thus, a nonzero state can have zero immediate cost. Observability ensures that it cannot remain invisible to this cost forever under zero input. When observability does not hold, an unstable trajectory may grow the state without appearing in the cost, so minimizing cost may not stabilize the system.